TimeRaw vs gaze/PSM peak features (position)
Positive signed lag = gaze leads; * BH q < 0.05 within each analysis panel

	Between participants			Within participant			Within participant adjusted for task order + attempt		
	Signed lag	Absolute lag	Peak r	Signed lag	Absolute lag	Peak r	Signed lag	Absolute lag	Peak r
A: Whole task (L)	+0.11	+0.07	-0.62	-0.41	-0.48	-0.00	-0.46	-0.35	+0.03
A: Whole task (R)	+0.00	+0.00	+0.70	-0.38	-0.38	-0.09	-0.02	-0.02	-0.09
A: Reach Ring (L)	-0.25	+0.05	-0.60	-0.03	+0.40	-0.25	-0.13	+0.43	-0.29
A: Reach Ring (R)	+0.28	-0.30	+0.42	-0.25	-0.40	-0.11	-0.53	-0.24	+0.09
A: Grab Ring (L)	+0.05	+0.27	-0.14	+0.04	+0.12	-0.09	+0.21	+0.13	-0.13
A: Grab Ring (R)	+0.41	-0.08	-0.24	-0.57	-0.22	-0.29	-0.30	-0.21	-0.48
A: Transfer Ring (L)	-0.08	+0.36	-0.55	-0.19	-0.39	-0.37	-0.10	-0.15	-0.49 *
A: Transfer Ring (R)	+0.72	+0.74	+0.29	-0.42	-0.33	+0.06	-0.08	-0.00	+0.08
A: Place Ring (L)	-0.05	+0.39	-0.15	+0.04	+0.04	-0.26	-0.13	+0.27	-0.16
A: Place Ring (R)	-0.19	-0.09	+0.22	+0.07	-0.36	+0.47	+0.33	-0.20	+0.49
A: Drop Ring (L)	excluded	excluded	excluded	excluded	excluded	excluded	excluded	excluded	excluded
A: Drop Ring (R)	excluded	excluded	excluded	excluded	excluded	excluded	excluded	excluded	excluded
C: Whole task (L)	+0.46	-0.15	+0.08	-0.37	-0.13	-0.12	-0.31	-0.25	+0.01
C: Whole task (R)	+0.06	-0.56	-0.27	+0.28	-0.11	-0.51 *	+0.30	+0.06	-0.55 *
C: Reach String (L)	+0.10	-0.05	+0.21	-0.13	-0.03	-0.46	-0.16	-0.02	-0.65 *
C: Reach String (R)	+0.32	+0.01	+0.06	+0.04	-0.51 *	-0.23	+0.12	-0.50	-0.25
C: Threading (L)	low N	low N	low N	low N	low N	low N	low N	low N	low N
C: Threading (R)	-0.31	+0.60	+0.49	low N	low N	low N	low N	low N	low N
C: Wrap around PSM (L)	+0.14	+0.25	-0.39	-0.45	-0.07	-0.37	-0.43	-0.11	-0.37
C: Wrap around PSM (R)	-0.43	+0.24	-0.94 *	-0.00	-0.28	-0.47	+0.14	-0.10	-0.38
C: Pull/cinch (L)	+0.15	-0.32	-0.09	-0.06	+0.08	-0.49	-0.21	+0.11	-0.49
C: Pull/cinch (R)	excluded	excluded	excluded	excluded	excluded	excluded	excluded	excluded	excluded

Rank association: 1.00, 0.75, 0.50, 0.25, 0.00, −0.25, −0.50, −0.75, −1.00

TimeRaw used directly, without sign reversal. Positive = longer duration; negative = shorter duration.
Available TimeRaw: A 33, C 33; 11 participants. Missing outcome values are excluded; phase-specific counts are in the CSV.

EcoRaw vs gaze/PSM peak features (position)

Positive signed lag = gaze leads; * BH q < 0.05 within each analysis panel

	Between participants			Within participant			Within participant adjusted for task order + attempt		
	Signed lag	Absolute lag	Peak r	Signed lag	Absolute lag	Peak r	Signed lag	Absolute lag	Peak r
A: Whole task (L)	-0.32	-0.41	-0.33	-0.16	-0.43	+0.17	-0.08	-0.28	+0.26
A: Whole task (R)	-0.39	-0.39	+0.19	-0.45	-0.45	-0.17	-0.17	-0.17	-0.21
A: Reach Ring (L)	-0.06	-0.41	-0.27	+0.00	+0.05	-0.20	-0.07	-0.02	-0.21
A: Reach Ring (R)	+0.23	-0.03	-0.31	-0.27	-0.44	-0.10	-0.51	-0.30	+0.08
A: Grab Ring (L)	+0.12	+0.16	-0.35	+0.46	+0.10	-0.24	+0.62	+0.09	-0.28
A: Grab Ring (R)	-0.11	+0.07	-0.42	-0.36	-0.16	-0.27	+0.00	-0.13	-0.42
A: Transfer Ring (L)	-0.21	-0.07	-0.42	-0.15	-0.36	-0.29	-0.05	-0.12	-0.34
A: Transfer Ring (R)	+0.59	+0.59	-0.55	-0.37	-0.37	-0.09	-0.05	-0.09	-0.14
A: Place Ring (L)	-0.02	+0.09	+0.19	-0.04	-0.23	-0.15	-0.20	-0.15	-0.02
A: Place Ring (R)	-0.28	-0.09	-0.26	+0.11	-0.24	+0.30	+0.34	-0.04	+0.23
A: Drop Ring (L)	excluded	excluded	excluded	excluded	excluded	excluded	excluded	excluded	excluded
A: Drop Ring (R)	excluded	excluded	excluded	excluded	excluded	excluded	excluded	excluded	excluded
C: Whole task (L)	+0.46	-0.13	-0.09	-0.41	-0.26	-0.00	-0.35	-0.39	+0.13
C: Whole task (R)	+0.07	+0.03	+0.10	+0.18	-0.12	-0.39	+0.17	+0.03	-0.41
C: Reach String (L)	+0.01	+0.14	+0.45	-0.05	-0.12	-0.45	-0.06	-0.13	-0.64
C: Reach String (R)	+0.14	+0.43	+0.17	-0.05	-0.54	-0.14	+0.00	-0.53	-0.15
C: Threading (L)	low N	low N	low N	low N	low N	low N	low N	low N	low N
C: Threading (R)	-0.94	+0.09	-0.14	low N	low N	low N	low N	low N	low N
C: Wrap around PSM (L)	+0.28	+0.70	-0.22	-0.35	-0.05	-0.28	-0.32	-0.08	-0.27
C: Wrap around PSM (R)	-0.21	+0.41	-0.68	+0.12	-0.27	-0.37	+0.28	-0.08	-0.27
C: Pull/cinch (L)	+0.16	-0.27	+0.18	+0.30	-0.21	+0.01	-0.16	-0.14	+0.06
C: Pull/cinch (R)	excluded	excluded	excluded	excluded	excluded	excluded	excluded	excluded	excluded

Rank association: 1.00, 0.75, 0.50, 0.25, 0.00, −0.25, −0.50, −0.75, −1.00

EcoRaw used directly, without sign reversal. Positive = larger raw motion measure; negative = smaller raw motion measure.
Available EcoRaw: A 33, C 33; 11 participants. Missing outcome values are excluded; phase-specific counts are in the CSV.